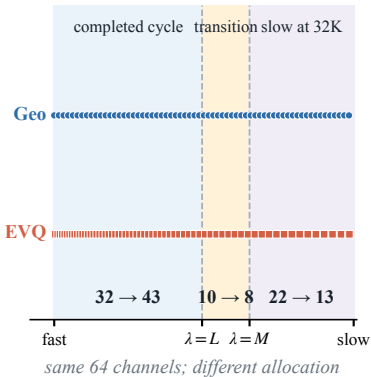
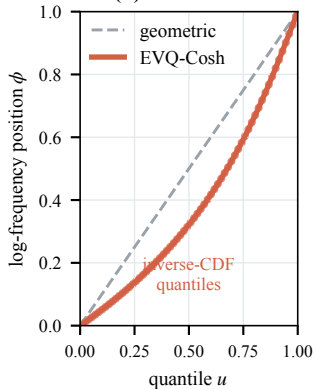


(a) A finite spectral budget



(b) Closed form



$$\phi_k = 1 - \tau^{-1} \operatorname{asinh}((1 - u_k) \sinh \tau)$$

(c) Two controls isolate allocation

FIX RANGE
same endpoints, span, init, data, budget
move only 30 interior frequencies
 $\Delta \text{OOD NLL} = -0.478 / -0.205 / -0.113$



MATCH PHASE EXPOSURE AT 1.485B
same Q/K adaptation and 4K/8K/16K phases
8K RULER: 2.02% → 31.63%

allocation remains binding under both controls